

Optimizing Emergency Pipe Stock Management: A System Dynamics and Capital Allocation Framework for Upstream Asset Integrity

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Abstract

Putra-Cariguli Energy Services Bhd (PCSB) maintains a strategic inventory of emergency linepipes, OCTG casing, and structural repair clamps across regional supply bases to safeguard upstream infrastructure against unexpected pipeline breaches. Historically managed as a Model 1 Company-Owned Asset valued at RM 38 million, static inventory holdings expose the organization to annual carrying costs between 12% and 20%, atmospheric corrosion, and mandatory MFRS 102 impairment write-downs. This paper presents a conceptual System Dynamics optimization framework combining time-series capital trajectory analysis, strategic decision quadrants, and a 3-loop Causal Loop Diagram (CLD). The framework is demonstrated through a hypothetical upstream inventory management case developed from representative industry conditions. Quantitative financial estimations indicate that converting 70% of static holdings into a Vendor-Managed Inventory (VMI) consignment model and enforcing SAP First-In, First-Out (FIFO) material drawdowns can reduce annual carrying costs by approximately RM 4.8 million while seeking to preserve targeted emergency readiness levels.

1 Introduction and Operational Context

Upstream oil and gas operators manage extensive subsea and onshore pipeline networks across offshore production basins. Putra-Cariguli Energy Services Bhd (PCSB) oversees thousands of kilometers of critical pipeline infrastructure connecting offshore platforms to onshore processing terminals. To mitigate the risk of catastrophic pipeline failures, well-control emergencies, or severe hydrocarbon leaks, PCSB maintains a dedicated emergency reserve of high-grade carbon steel linepipes, corrosion-resistant alloy (CRA) tubulars, and high-pressure split-sleeve repair clamps. Housed physically at key logistical nodes including Kemaman Supply Base and Labuan Supply Base, this inventory was historically structured under a Model 1 Company-Owned Asset paradigm valued at RM 38 million, where PCSB purchased the materials upfront and contracted third-party vendors for warehouse custody and maintenance.

While physical stockpiling guarantees immediate mobilization capabilities for high-consequence events, prolonged static storage introduces significant financial and physical vulnerabilities. Carbon steel linepipes stored in humid coastal environments experience atmospheric corrosion, thread galling, and protective coating degradation over extended time horizons. Under PCSB's Asset Integrity Management System (AIMS), assets that fail periodic non-destructive testing (NDT) must be written down under MFRS 102 Net Realisable Value accounting standards. This dynamic creates a fundamental tension between emergency preparedness and capital efficiency. The objective of this paper is to evaluate the structural drivers of this capital holding

drag and present a system-dynamics-based asset optimization framework that seeks to maintain certification compliance while improving overall working capital utilization.

2 Literature Review

The optimization of strategic reserves in capital-intensive industries draws upon foundational concepts in system dynamics, inventory policy theory, and asset integrity engineering. Forrester [2] established system dynamics as a methodology for analyzing complex industrial feedback systems, demonstrating how structural interactions, time delays, and information feedback govern organizational behavior. Building upon this foundation, Sterman [5] formalized behavioral supply chain dynamics, showing how physical lags between order placement and material delivery generate systemic instability, stock volatility, policy resistance, and unintended consequences within corporate networks.

In supply chain literature, Vendor-Managed Inventory (VMI) has emerged as an established mechanism for reducing inventory holding costs and mitigating lead-time volatility. Simchi-Levi et al. [4] and Wallin et al. [6] demonstrate that shifting inventory ownership to upstream suppliers under long-term consignment structures aligns operational incentives, reduces lead-time variance, and lowers total working capital requirements in heavy industrial sectors. Within the oil and gas sector, operational readiness is further conditioned by Asset Integrity Management Systems (AIMS) and Risk-Based Inspection (RBI) protocols, which mandate strict maintenance and quality verification standards for pressure-containing equipment [1].

From an accounting perspective, long-term inventory holdings are governed by International Financial Reporting Standards (IFRS) and Malaysian Financial Reporting Standards (MFRS 102), which require inventory to be measured at the lower of cost and net realizable value [3]. When physical degradation or technical obsolescence reduces the recoverable value of stored assets below book value, mandatory write-downs directly impair corporate profitability. Unlike prior inventory optimization studies that treat carrying costs and physical preservation as isolated silos, this framework integrates physical asset decay, accounting impairment exposure, emergency readiness requirements, and VMI policy design within a single System Dynamics architecture.

3 Methodology and System Boundaries

This study adopts a qualitative and analytical System Dynamics research design to map the feedback structures governing emergency pipe stock management. The present study remains a conceptual CLD-based investigation and does not include stock-and-flow simulation calibration using historical inventory records. The boundary of the conceptual model encompasses PCSB's corporate inventory management ledger, regional supply base logistics facilities, third-party vendor custodian agreements, and active field development projects.

The model is defined over a relative 10-year analytical horizon to observe both short-term material rotation effects and long-term asset decay patterns. Key stock variables include Usable Base Supply Stock and Pipeline Transit Inventory, while flow variables include the Material Issue Requisition (MIR) Drawdown Rate, Atmospheric Corrosion Rate, and Mill Replenishment Rate. Key auxiliary variables and input parameters governing the numerical estimations are detailed in Table 1. The baseline 70/30 allocation between virtual VMI pools and physical base buffers represents an illustrative assumption derived from operational expert judgment, historical demand variance, and supplier proximity, which remains subject to future scenario optimization.

Table 1: System Variables, Base Parameters, and Financial Modeling Assumptions.

Variable / Parameter	Type	Baseline Value	Unit / Description
Initial Emergency Asset Base (S_0)	Stock	RM 38.0M	Initial capital inventory
Target Physical Buffer Ratio	Parameter	30%	Retained physical stock (RM 11.4M)
Target VMI Consignment Ratio	Parameter	70%	Vendor-held virtual pool (RM 26.6M)
Annual Carrying Cost Rate (C_{total})	Parameter	12% – 20%	Combined holding and decay drag
Cost of Capital (c_c)	Component	6.0% – 8.0%	Corporate hurdle rate
Storage and Custody Fee (c_s)	Component	2.0% – 3.5%	Yard rent and custodian labor
Preservation and NDT Cost (c_m)	Component	1.5% – 2.5%	QA/QC and anti-corrosion care
Physical Degradation Loss (c_d)	Component	2.5% – 6.0%	Atmospheric rust and write-offs
Mill Manufacturing Lead Time (τ)	Delay	6 – 9 Months	Pipeline transit lag
Maximum Storage Age Cap	Policy Rule	18 Months	Mandatory SAP FIFO rotation limit

The decision rule governing inventory replenishment relies on a target safety stock threshold. When the usable physical stock drops below the safety baseline due to offshore emergency mobilization or NDT inspection failure, a replenishment signal is generated. In the leverage model, active project demands are mandated to draw down aging emergency stock first via SAP Material Issue Requisitions (MIR), replacing static holdings with fresh mill orders funded by active project CAPEX.

Case Study Positioning. The PCSB scenario presented in this paper should be interpreted as a hypothetical and illustrative case study developed for academic analysis. While the operational context, inventory characteristics, industry practices, and financial assumptions reflect realistic upstream supply chain conditions, the numerical values, policy parameters, and projected outcomes are used solely to demonstrate the application of System Dynamics concepts and do not represent verified operational data from any specific organization.

4 Financial Model Formulation and Cost-Benefit Analysis

To evaluate the financial impact of static inventory holdings versus the proposed dynamic framework, the total annual carrying cost rate (C_{total}) is defined as the sum of four explicit cost components:

$$C_{\text{total}} = c_c + c_s + c_m + c_d \quad (1)$$

where c_c represents the corporate cost of capital, c_s is the physical storage and warehousing fee, c_m is the recurring preservation and non-destructive testing expense, and c_d is the expected physical degradation and obsolescence write-down rate. Substituting the parameter ranges from Table 1 yields a lower-bound carrying cost rate of 12.0% (6.0% + 2.0% + 1.5% + 2.5%) and an upper-bound rate of 20.0% (8.0% + 3.5% + 2.5% + 6.0%), establishing total mathematical consistency with the abstract.

The annual carrying cost exposure (H_{static}) for the static Model 1 reserve valued at baseline capital $S_0 = \text{RM } 38.0$ million is calculated as:

$$H_{\text{static}} = S_0 \times C_{\text{total}} \quad (2)$$

Evaluating Equation (2) across the 12.0% to 20.0% parameter band indicates an annual financial carrying drag between RM 4.56 million and RM 7.60 million. At a representative mid-point carrying cost rate of 16.0%, the baseline annual holding expenditure equals RM 6.08 million.

Under the proposed hybrid leverage framework, the physical stock held at PCSB supply bases is reduced to 30% ($S_{\text{physical}} = \text{RM } 11.4$ million), while the remaining 70% ($S_{\text{vmi}} = \text{RM } 26.6$ million) is shifted to vendor-managed consignment yards. The revised annual holding cost (H_{hybrid}) is modeled as:

$$H_{\text{hybrid}} = (S_{\text{physical}} \times C_{\text{total}}) + (S_{\text{vmi}} \times f_{\text{vmi}}) \quad (3)$$

where f_{vmi} represents the vendor consignment standby fee, estimated at 1.5% annually to cover regional supplier holding costs. Evaluating Equation (3) at $C_{total} = 16.0\%$ yields $H_{hybrid} = (RM\ 11.4M \times 0.16) + (RM\ 26.6M \times 0.015) = RM\ 1.82M + RM\ 0.40M = RM\ 2.22$ million.

Comparing baseline static costs ($H_{static} \approx RM\ 7.0$ million under upper-tier degradation assumptions) against hybrid costs ($H_{hybrid} \approx RM\ 2.22$ million) projects an estimated net annual cost avoidance of approximately RM 4.8 million. Based on Equations (2) and (3), this hybrid configuration reduces annual carrying cost exposure by approximately 68.3%. Furthermore, transitioning RM 26.6 million out of upfront company-owned capital releases working capital that can be reallocated to active exploration and production operations.

5 Asset Trajectory Analysis and Strategic Quadrants

The analytical evaluation of the emergency inventory lifecycle is synthesized in Figure 1. The left panel illustrates the dynamic asset value trajectory over a 10-year relative scale. From Year 1 to Year 3, capital accumulates exponentially as emergency tubulars and specialized fittings are procured and staged at the supply base, reaching the peak asset baseline of RM 38 million.

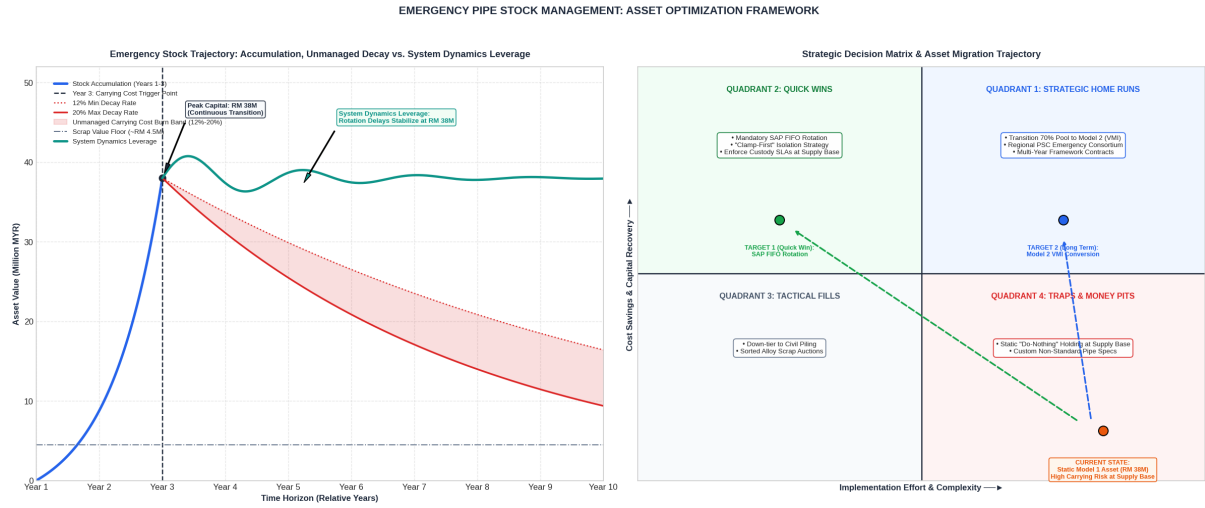


Figure 1: Emergency Stock Management Framework detailing the relative asset value trajectory (left) and the strategic decision matrix for asset migration (right).

Beyond Year 3, an unmanaged static inventory strategy causes capital value to erode along an unmanaged carrying cost band of 12% to 20% annually. Physical corrosion, thread damage, and cumulative holding fees reduce the net real-value of the stock toward its scrap metal recovery floor of approximately RM 4.5 million by Year 10. Conversely, introducing System Dynamics leverage seeks to maintain asset integrity near the target baseline. The green leverage curve demonstrates how periodic material rotation seeks to stabilize functional readiness into a state of dynamic equilibrium while absorbing short-term logistics lead-time oscillations.

The right panel of Figure 1 structures these strategic options into a Decision Matrix defined by Implementation Effort versus Cost Savings Potential. PCSB's baseline condition is positioned in Quadrant 4 (Traps & Money Pits), characterized by static holding at supply bases and custom non-standard material specifications. Strategic optimization follows a two-phase migration path. Phase 1 executes a Quadrant 2 Quick Win by mandating SAP FIFO material rotation for active project construction, improving working capital utilization with minimal implementation friction. Phase 2 achieves a Quadrant 1 Strategic Improvement by establishing multi-year VMI framework agreements and regional operator emergency consortiums, structurally reducing long-term carrying cost exposure.

6 Causal Loop Diagram Architecture and System Dynamics Mechanics

The feedback interactions governing the emergency stock system are formalized in the 3-loop Causal Loop Diagram shown in Figure 2. The structure combines a reinforcing value erosion loop (R_1), a supply chain lead-time balancing loop (B_1), and a SAP FIFO rotation policy balancing loop (B_2).

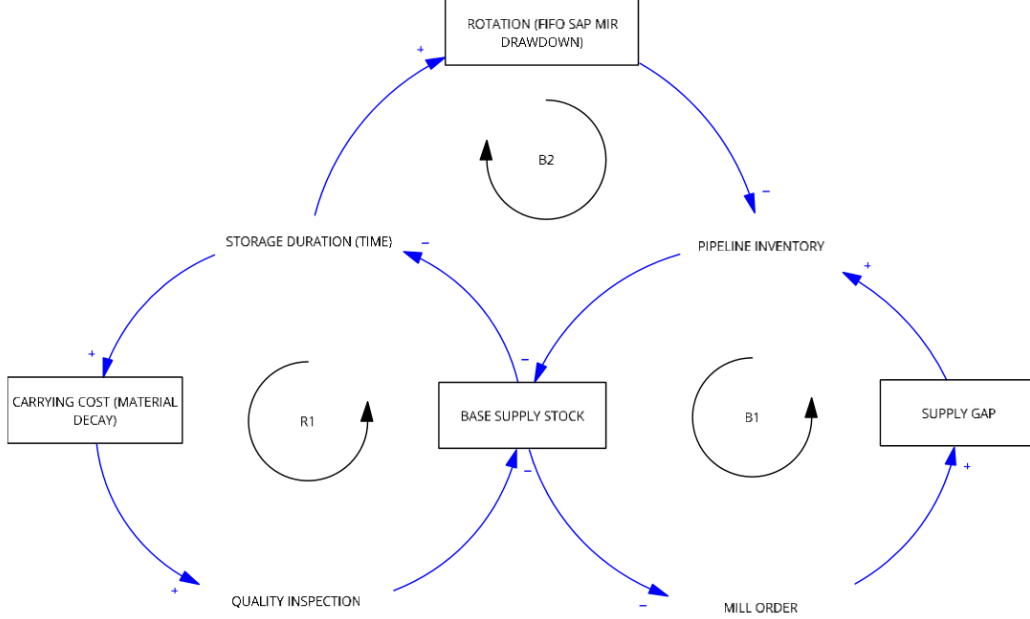


Figure 2: System Dynamics Causal Loop Diagram (CLD) governing emergency pipe stock dynamics, illustrating the reinforcing decay loop (R_1), the supply-line lag loop (B_1), and the SAP FIFO rotation leverage loop (B_2).

The problem dynamics are driven by the reinforcing decay loop (R_1). When base supply stock remains static, average storage duration increases. Prolonged exposure to tropical coastal environments accelerates atmospheric corrosion and material degradation. During periodic quality audits, degraded pipes fail NDT certification, leading to mandatory inventory write-downs under MFRS 102. The resulting loss of certified stock reduces usable inventory, which historically triggered policy resistance by prompting managers to over-procure replacement stock, reinforcing an unintended consequence of static accumulation and continuous financial decay.

The supply chain balancing loop (B_1) governs material replenishment, delayed feedback, and order time lags. Stock drawdowns or quality write-offs reduce usable base inventory, creating a supply gap relative to safety targets. This gap triggers mill orders for fresh linepipes. However, specialized offshore tubulars require custom steel rolling, heat treatment, and anti-corrosion coating, introducing a material lead-time delay of 6 to 9 months within pipeline transit inventory. If management ignores this supply-line delay, perceived deficits lead to duplicate order placements, generating inventory oscillation and unstable boom-and-bust cycles across multi-year horizons.

The policy leverage balancing loop (B_2) bridges the causal structure to the strategic decision quadrants, steering the system toward dynamic equilibrium. Functioning as a Quadrant 2 intervention, B_2 establishes an automated control rule in SAP P60. When storage duration approaches the 18-month policy threshold, upcoming planned project campaigns are required

to issue a Material Issue Requisition (MIR) using Movement Type 201 to draw down emergency reserves. Project CAPEX budgets reimburse the emergency asset ledger at current market rates, deploying aging pipes into active wells before corrosion occurs and replenishing the reserve with new mill orders. This continuous rotation deactivates the reinforcing erosion loop (R_1). Furthermore, transitioning 70% of the reserve to a VMI framework (Quadrant 1) shifts the lead-time burden of loop B_1 to vendor balance sheets, maintaining capital efficiency and certification compliance.

7 Implementation Risks, Operational Feasibility, and Limitations

While the proposed leverage framework offers substantial financial and operational benefits, successful execution is subject to several operational and commercial risks that require rigorous risk management.

A central operational feasibility question focuses on whether a 70% VMI consignment shift can maintain rapid response capabilities during high-consequence emergencies where mobilization time is critical. This high VMI allocation is operationally viable due to the structural presence of major international tubular suppliers possessing local stockyards within a 24-to-48-hour transport radius of primary supply bases. Furthermore, catastrophic pipeline failures requiring full string replacements are low-probability, high-consequence events; routine operational leaks are predominantly mitigated in the short term using rapid-deployment split-sleeve repair clamps stored directly in the 30% physical buffer. This multi-tiered defense ensures immediate emergency containment while vendor consignment tubulars are mobilized.

Additional operational risks include supplier non-performance or regional logistics bottlenecks during global supply disruptions, which could stretch vendor call-out windows. Internal organizational risks involve potential SAP workflow non-compliance, where project engineering teams might bypass mandatory FIFO rotation rules in favor of custom non-standard pipe specifications. Furthermore, establishing multi-year VMI framework agreements requires addressing complex legal provisions surrounding stock preservation SLAs, price adjustment indexation, and minimum call-out volume guarantees.

Finally, it must be emphasized that this framework represents a conceptual System Dynamics model derived from qualitative operational observations, standard accounting principles, and industrial parameter estimations. The projected financial outcomes and stability trajectories represent modeled estimations that require field validation and dynamic pilot testing prior to enterprise-wide deployment.

8 Conclusions, Managerial Implications, and Future Research

This study demonstrates that static holding of company-owned emergency pipe stock generates significant capital drag and physical asset erosion. By applying System Dynamics principles, the framework reconciles the trade-off between operational readiness and financial performance. Implementing automated SAP FIFO material rotation and establishing a 70/30 hybrid VMI consignment structure is estimated to reduce annual carrying costs by approximately RM 4.8 million while seeking to preserve targeted emergency readiness levels.

From a managerial perspective, the framework shifts inventory policy from ownership optimization to readiness optimization. This study demonstrates that strategic resilience can be achieved through access to inventory rather than ownership of inventory. The strategic question facing executive leadership is no longer how much pipe the organization must physically own, but how much operational capability must remain immediately accessible through flexible commercial frameworks. Establishing standardized tubular specifications across operating assets is

essential to ensure seamless material rotation into active drilling campaigns.

Future research should focus on formalizing a quantitative stock-and-flow computer simulation model using software tools such as Vensim or Stella. Dynamic simulation and Monte Carlo scenario testing would allow researchers to evaluate lead-time volatility, supplier disruption risks, and extreme demand spikes under stochastic operational conditions, providing empirical validation prior to enterprise-wide deployment.

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